(1) 1.

Dokažte:

$$\forall x \in \mathbb{N} : \ \sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$$

(2) 2.

Určete $\sup, \inf, \max, \min$ množiny M kde:

$$M = \left\{ (-1)^n + \frac{1}{n^2+1} \mid n \in \mathbb{N} \right\}$$

(3) 3.

Spočtěte limitu:

$$\lim_{x \to 0} \frac{\sqrt[3]{1+x} - 1}{x^2 + x}$$